

Toward Image-to-Tactile Cross-Modal Perception for Visually Impaired People

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Abstract—It is still a great challenge for the visually impaired people to perceive their surroundings from a global perspective, which makes it difficult for them to interact with unfamiliar environments. The reason is that these conventional assisting devices only address the obstacle avoidance problem. They do not provide visually impaired people with a global perception of the surrounding environment. In this article, a new generative adversarial network (GAN) model is developed to effectively transform the ground images into the tactile signal, which can be displayed by an off-the-shelf vibration device. The algorithm module and the hardware are integrated into a portable device, which provides visually impaired people with effective surrounding perception capability. In addition, a visual-tactile cross-modal data set is constructed to train the proposed deep-learning architecture. Experimental results show that the proposed system can help visually impaired people sense the ground and bring a better traveling experience for them.

Note to Practitioners—This article presents a portable device that provides tactile recognition assistance for visually impaired people. Such a technology can be extensively used in tactile mouse and white cane. The developed technology can be extensively used for various industrial applications, such as surrounding monitoring and manipulation. The proposed work demonstrates the promising ability of artificial intelligence in healthcare applications. The generated tactile signals are expected to be used in many human-centered systems, and we believe that our contribution is an important step toward the development of a more comprehensive assisting technology for visually impaired people.

Index Terms—Cross-modal perception, generative adversarial networks (GANs), visually impaired.

I. INTRODUCTION

VISUAL impairment has a significantly negative influence on people's quality of life. It limits their daily functioning and mobility and even affects their mental health. The World

Health Organization once estimates that about 285 million people around the world are with visual impairment, among which 39 million are blind. One of the main causes of visual impairment is related to elderly diseases, and the world aging population is growing quicker than ever before. We can expect that more people will suffer from visual impairment in the coming years. Therefore, it is urgent to develop novel effective assistive tools to help them live safely and independently [1]–[4].

Some assistive devices or companions, such as white cane and guide dogs, are usually used to guide visually impaired people to move around safely in normal lives [5]. They are able to provide a variety of feedback signals, containing acoustic, tactile, and so on to assist visually impaired people [6], [7]. Islam *et al.* [8] describe the state-of-the-art technologies on walking assistive tools for visually impaired people. Products such as UltraCane and TomPouce have already been commercialized in practical applications [9], which can provide the location of remote overhanging obstacles.

Among various feedback signals provided by the assistance devices, the tactile modality, which is very important for physical human–robot interaction [10]–[12], is especially preferred due to its intuition. Research is dedicated to provide the user with a proper tactile stimulus that can reflect the environment perceived through touch and kinesthesia, e.g. body position, movement, and weight [13], [14]. Andò *et al.* [15] introduce a short cane, which includes an active handle and is embedded with a smart sensing strategy. There is a haptic tool installed in the cane handle, and it can generate a tactile sensation to a person's palm with the designed codification strategy. The tactile signals can be used as a warning when obstacles are around. O'Modhrain *et al.* [16] demonstrate how useful and important the haptics is for visually impaired people to acquire information.

However, the current assistive devices and vision techniques only focus on providing notices to the visually impaired people when obstacles are detected. It is still a great challenge for the visually impaired people to perceive their surroundings from a global perspective, which makes it difficult for them to interact with unfamiliar environments [17].

On the other hand, due to the development of artificial intelligence and machine-learning technologies, a lot of cross-modal learning approaches are proposed to investigate the connection between different sensing modalities [18]–[21]. Specifically, generative adversarial networks (GANs) [22], which used two deep networks to contest with each other,

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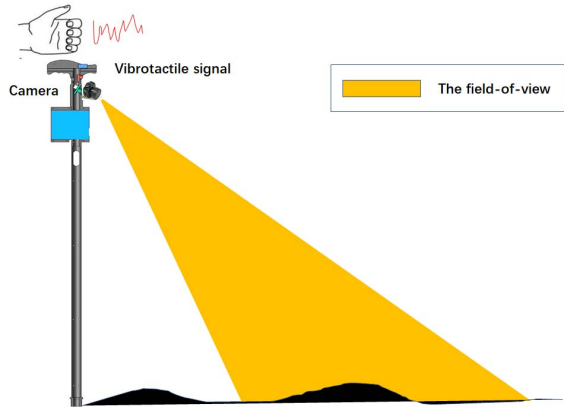


Fig. 1. Principle of the image-to-tactile cross-modal transformer.

have shown very promising results to generate new sample images [23], texts [24], speeches, and so on.

Motivated by the great success of GAN to generate a new sample for cross-modal perception, we have developed a new portable device for visually impaired people in this article (see Fig. 1). The main contributions are summarized as follows.

- 1) A new GAN model is developed to effectively transform the ground images into the tactile signal, which can be displayed by an off-the-shelf vibration device.
- 2) A visual-tactile cross-modal data set is constructed to train the proposed deep-learning architecture.
- 3) The algorithm module and the hardware are integrated into a portable device, which provides visually impaired people with effective surrounding perception capability.

The proposed work demonstrates the promising ability of artificial intelligence in healthcare applications. Meanwhile, the developed technology can be extensively used for various industrial applications, such as surrounding monitoring and manipulation. Also, with generated tactile signals directly provided to the human, it is expected to be used in many human-center systems.

The rest of this article is organized as follows. The related work is reviewed in Section II. The proposed network architecture is described in Section III. Section IV describes the construction of the data set in detail. The experimental results are presented in Section VI. Finally, Section VII presents the user study.

II. RELATED WORK

Tactile display has been extensively studied for several decades, and many prototypes have been developed [25]–[29]. Here, we focus on the visual-tactile related work and divide them into the following two categories.

A. Tactile Signal Retrieval From Images

Given an image, an existing tactile signal is selected from a beforehand constructed gallery. Yuan *et al.* [30] capture the RGBD images and tactile data of draped fabrics, and a CNN network is jointly trained across different modalities of data. After training, for each input, regardless of the modality, the CNN network will generate an embedding vector,

which is supposed to reflect the fabric's physical property. By comparing the generated embeddings, the proposed system is able to predict how a fabric feels given its image, and vice versa. Similarly, Li *et al.* [31] introduce new tasks of synthesizing plausible tactile signals from visual inputs as well as imagining how we interact with objects given tactile data as input. Liu *et al.* [32], [33] established a sparse coding and dictionary learning method for retrieval tactile sequences given an image query. Recently, Takahashi and Tan [34] provide a new method to analyze the tactile properties from images.

B. Tactile Signal Generation From Images

Compared with the above-mentioned topic, the tactile signal generation from images is more challenging because it is supposed to produce new tactile signals. Ujitoko and Ban [35] use GANs to simulate the distribution of time series data and indirectly generate vibration signals from images through time–frequency conversion. Recently, Li *et al.* [36] propose a method to identify the categories of input images and then output the corresponding tactile spectrum. However, this method will lose a lot of high-dimension feature information by classifying images as category labels.

The work proposed in this article belongs to the category of Tactile Generation from Images. The main difference between our work and the work in [35] and [36] is that we propose new learning models and develop a portable device for the visually impaired people.

III. DEEP-LEARNING ARCHITECTURE

Motivated by the work of [35], we represent the tactile signals using the amplitude spectrogram and utilize the convolutional neural networks to extract the feature vectors. Therefore, the main task of the proposed work is to automatically produce the spectrum maps whose features are determined by the target ground. In this regard, the visual-tactile cross-modal transformation is formulated as an image-to-image translation problem, which can be solved using the popular GANs [37], [38].

In order to transform the ground image into spectrums without any intervention from the users, we adopt the cross-domain GAN as the basic part of this method. We build an end-to-end generation network, and our method can recognize the target image without adding additional classification structure. In this article, we adopt the DiscoGAN [38] to translate the ground images into the spectrogram domain. The whole deep-learning architecture is illustrated in Fig. 2.

A. DiscoGAN for Image-to-Tactile Transformation

For given domains X and Y , the generators G and F transfer an image from X to Y or Y to X , respectively. Also, the discriminators D_X and D_Y determine if a sample belongs to the domain X or Y , respectively. The target is to determine the parameters in the neural networks G , F , D_X , and D_Y .

First, we define two reconstruction terms for G and F as

$$\mathcal{L}_{\text{rec}}(G) = \mathbb{E}_{x \sim P_{\text{data}(x)}} [\|F(G(x)) - x\|_1] \quad (1)$$

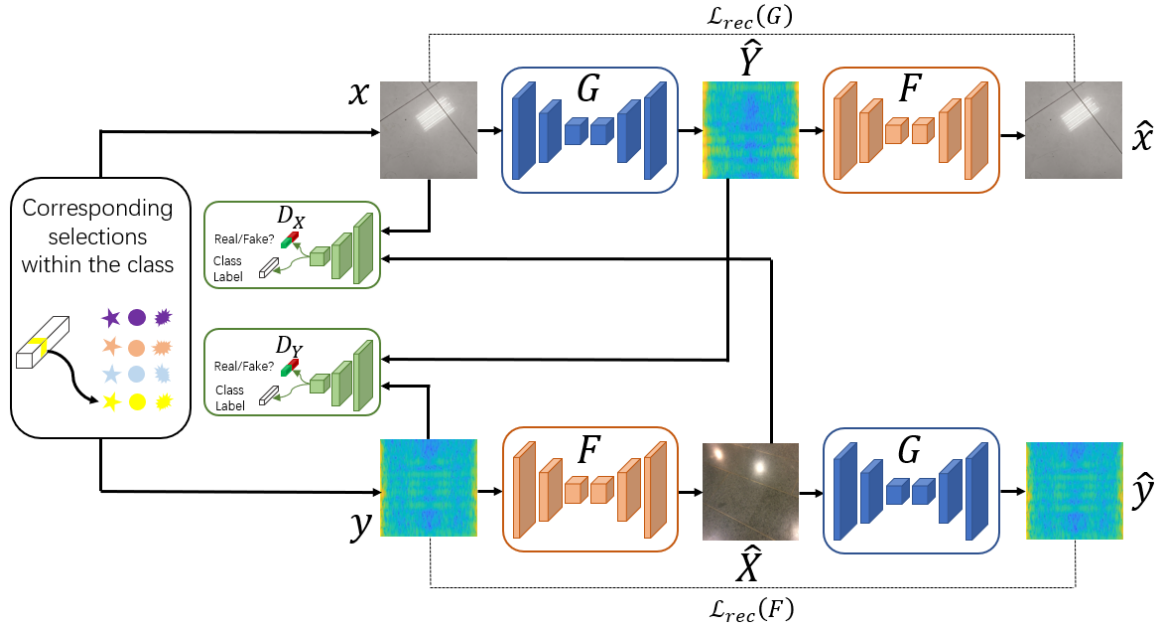


Fig. 2. Framework of our proposed structure. Our model relies on DiscoGAN to learning the translation between the distributions of ground images and spectrograms. In order to double constrain the generated results by using the trained D_X and D_Y as recognizers, an additional auxiliary classification layer is added to the original discriminator. During the training, the network will automatically select the corresponding spectrum (or ground image) according to the category of the ground image (or spectrogram), thus ensuring the interclass difference and intraclass similarity of the generated images.

and

$$\mathcal{L}_{\text{rec}}(F) = \mathbb{E}_{y \sim P_{\text{data}}(y)} [\|G(F(y)) - y\|_1] \quad (2)$$

where $P_{\text{data}}(x)$ and $P_{\text{data}}(y)$ are the data distribution over domains X and Y , respectively. Such terms use L_1 penalty to encourage the generators G and F to be reconstructed by the original samples and establish a strong mapping between the domains.

Furthermore, we define the adversarial loss terms as

$$\begin{aligned} \mathcal{L}_{\text{adv}}(D_Y) = & -\mathbb{E}_{y \sim P_{\text{corry}}(x)} [\log D_Y(y)] \\ & -\mathbb{E}_{x \sim P_{\text{data}}(x)} [\log(1 - D_Y(G(x)))] \end{aligned} \quad (3)$$

and

$$\mathcal{L}_{\text{adv}}(G) = -\mathbb{E}_{x \sim P_{\text{data}}(x)} [\log D_Y(G(x))] \quad (4)$$

where $P_{\text{corry}}(x)$ represents the selection of y according to the corresponding categories of images. The goal of G is to generate new samples which confuse the discriminator D_Y , while the goal of D_Y is to discriminate the true and fake samples as correct as possible.

Similarly, we have

$$\begin{aligned} \mathcal{L}_{\text{adv}}(D_X) = & -\mathbb{E}_{x \sim P_{\text{corry}}(y)} [\log D_X(x)] \\ & -\mathbb{E}_{y \sim P_{\text{data}}(y)} [\log(1 - D_X(F(y)))] \end{aligned} \quad (5)$$

and

$$\mathcal{L}_{\text{adv}}(F) = -\mathbb{E}_{y \sim P_{\text{data}}(y)} [\log D_X(F(y))] \quad (6)$$

where $P_{\text{corry}}(y)$ represents the selection of x according to the corresponding categories of tactile samples.

The above-mentioned cycle generation network can only realize domain transformation and cannot reflect differences

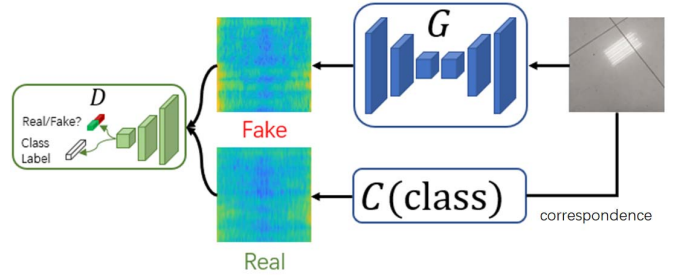


Fig. 3. We input the ground image to the generator, and the network automatically selects one of the real spectrum of the corresponding category as the input of the discriminator.

between categories. Previous work only used labels or images for the conditions of GAN. To be suitable for our application scenarios, we improve the input method of conditions. When training G , we use images as the conditions and use the corresponding category labels of images as the conditions of D (see the details in Fig. 3).

Motivated by the design of ACGAN, we add a classification loss to the generator to increase the gap between the categories of the generated image, thus making the spectrogram more recognizable. We set an auxiliary classification layer in the discriminator, which can not only improve the quality of generated images but also enable the discriminator to be used as an independent classifier after the training of the overall network. To this end, we design the following classification terms:

$$\begin{aligned} \mathcal{L}_{\text{cls}}(D_Y) = & \mathbb{E}_{y \sim P_{\text{corry}}(x)} [\log P[\text{label}_y | y]] \\ & + \mathbb{E}_{x \sim P_{\text{data}}(x)} [\log P[c | D_Y(G(x))]] \end{aligned} \quad (7)$$

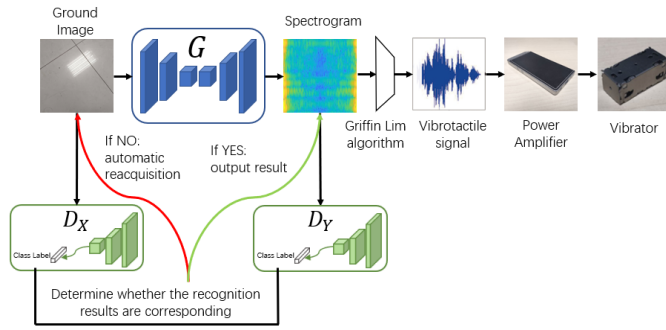


Fig. 4. Tactile signal generation.

and

$$\mathcal{L}_{\text{cls}}(G) = \mathbb{E}_{x \sim P_{\text{data}(x)}} [\log P[c|D_Y(G(x))]] \quad (8)$$

where c is the category of the sample x . Similarly, we can define the terms $\mathcal{L}_{\text{cls}}(D_Y)$ and $\mathcal{L}_{\text{cls}}(F)$. The whole objective function can be formulated as

$$\begin{aligned} \mathcal{L}(G, F) &= \mathcal{L}(G) + \mathcal{L}(F) \\ &= \mathcal{L}_{\text{adv}}(G) + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}(G) + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(G) + \mathcal{L}_{\text{adv}}(F) \\ &\quad + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}}(F) + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(F) \end{aligned} \quad (9)$$

and

$$\begin{aligned} \mathcal{L}(D_X, D_Y) &= \mathcal{L}(D_X) + \mathcal{L}(D_Y) \\ &= \mathcal{L}_{\text{adv}}(D_X) + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(D_X) + \mathcal{L}_{\text{adv}}(D_Y) \\ &\quad + \lambda_{\text{cls}} \mathcal{L}_{\text{cls}}(D_Y) \end{aligned} \quad (10)$$

and the optimization problem becomes

$$\min_{G, F} \max_{D_X, D_Y} \mathcal{L}(G, F) + \mathcal{L}(D_X, D_Y) \quad (11)$$

where λ_{rec} and λ_{cls} are the balancing factors for the reconstruction loss and classification, respectively. We adjust them to control the importance of each part. Those loss functions ensure that the network can generate near-real images with good generalization within classes and large differences between classes. By optimizing this objective function, we can disclose the complex correlations between ground images and their associated spectrograms.

B. Tactile Signal Generation

The tactile signal generation procedure is shown in Fig. 4. When the ground image is captured, D_X will classify it and output a label, and G will transform it into the spectrogram, which will also output a classification result through D_Y first. We stipulate that if the category labels identified by D_X and D_Y are consistent, the generated result is considered to be valid, and the spectrum can be generated. Otherwise, it is deemed to be invalid, and the network will select the next image to generate. The experimental results presented in Section V show this method's effectiveness. Finally, we utilize the Griffin-Lim algorithm [39] to convert the spectrogram into the vibration wave, which is transmitted to the user through amplifier and vibrator.

In our implementation, we use TensorFlow Lite, which is a lightweight solution of TensorFlow for mobile devices. It allows users to run deep-learning models on mobile devices with low latency, so users can use it for classification, regression, or target detection without having to interact with the server.

IV. DATA SET

For training the deep neural network, we construct a novel visual-tactile cross-modal data set, which is named ground-image-vibration (GroImgVib), which consists of seven common grounds: asphalt (A), outdoor slab (O), interior marble (I), wood floor (W), rubber track (R), and carpet and grass (C).

A. Ground Image

The ground photographs are taken by the camera integrated into our structural model. This ensures that the quality of images used for training is consistent with those collected in actual use. We collect seven kinds of ground under the conditions of sufficient light and low light, respectively. We take 100 pictures of each light intensity; therefore, there were 1400 digital camera images in the data set (see the top panel of Fig. 5).

Considering that different phones produce high-definition pictures in different sizes, and it is not necessary to use such high-resolution images to complete our task, so we need to unify the collected images to a smaller size. We crop the 1400 images and resized them into 128×128 size. The reduction of the image is not recommended to be simply reduced in size, which will cause serious distortion. Thus, we use the area interpolation method provided in TensorFlow to scale the image.

B. Vibration Image

Visual images are easy to capture and the information they contain is also obvious. However, the vibration information cannot be described intuitively. Considering the feasibility of the vibration signal generating in [35], we choose acceleration as the medium. To facilitate the processing of these data, we convert the acceleration time series signals to images.

We use the acceleration signals to be the vibrotactile stimulus. The establishment of the data set refers to the LMT-108-Surface Materials database [40], which includes various types of texture images and acceleration signals generated by corresponding tools for sliding tasks. Each type has 20 sets of acceleration signal samples, and each set contains three time-series signals of X , Y , and Z . All of the samples are annotated with 108 categories.

We select the seven-class subset from 108 categories that can correspond to the investigated ground. Each signal is 4 s long, and the sampling rate is set as 10 kHz. The 512-point short-time Fourier transform (STFT) with a 512-Hamming window and a 128-hop size is used to compute the spectrogram, and we convert the linear amplitude of the spectrogram to the logarithmic scale. We process the spectrogram to the same 128×128 size as the ground image. Therefore,

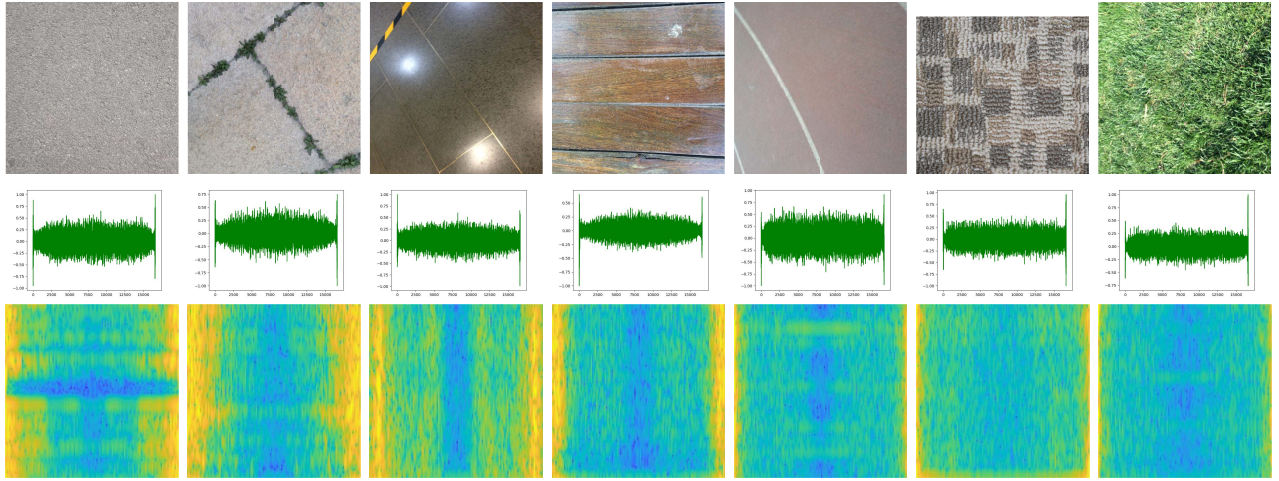


Fig. 5. Top: example images of different ground types in the data set. Middle: tactile signals. Bottom: representation of the vibration images.

the obtained spectrogram includes the information of the time–frequency domain up to 256 Hz for 1.625-s long. The values in the spectrogram are normalized into the range from 0 to 1. As a result, there are 140 amplitude spectrums in the data set (see the bottom of Fig. 5).

V. EXPERIMENTAL RESULTS

A. Data Preprocessing

Although our data set has 100 photographs for each class of ground, it is still not enough for training a neural network, so we follow the data augmentation settings when training the model. We flip each image horizontally and vertically, rotate them by a random angle, and use random erasing and mix-up techniques in addition to the traditional methods. We expand the ground data for each category to 1000, so we have a total of 7000 ground images with the full size 128×128 . The spectrogram cannot be used for data enhancement because they represent the time series characteristics and amplitude strength of the vibration signal, and similar processing methods will seriously affect the hidden information. We label both the ground images and the spectrograms, so that the corresponding data can be selected according to the category information when training the model, and this is also a necessary step to train a network with recognition function.

B. Training Details

According to the detailed analysis of the gradient in the condition of ACGAN in [41], we find that the model achieves the best performance when the value of λ_{cls} is the inverse of the number of categories, so we set λ_{cls} to 0.14. By verifying the experience of [37], we set λ_{rec} to 10. All models optimized use Adam optimizer [42]. We use a batch size of 8 in the training procedure. The learning rate is initialized to 0.0002 and exponentially decreases after 100 iterations of training.

To evaluate the feasibility of the proposed method, we trained an auxiliary classification generation network based on DiscoGAN. Training our model is done with a desktop PC

equipped with Intel Core i7-8750H CPU, 16-GB DDR4 RAM, and NVidia Geforce GTX 1080 Ti.

C. Generated Results

It can be seen from the experimental results that our method can generate corresponding categories of spectrograms according to the ground inputs, and the generating effect can ensure a good diversity without losing strong similarity. According to the results generated by different epochs, it can be seen that the images generated after 10 epochs have considerable similarity with the real spectrum, but the details are fuzzy and the sense of granularity is strong. After 200 epochs of training, the results have certain clarity, while after 400 epochs of training, the generated spectrograms have high similarities with the ground truth. Thus, we stop the network training after 400 epochs (see Fig. 6 for the detailed illustrations).

In the real scene, it is difficult to ensure the ground is clear. Those sundry objects constitute a disturbance to our ground recognition. However, all the pictures that we used are clear-ground data to train the neural network. In order to test the generalization of the model, our tests are divided into clear-ground test and messy ground test. For each ground condition, we select five different areas for experiments and conducted 20 clear ground and 20 messy ground tests in each area, so a total of 1400 experimental results are collected.

The clear-ground images are misidentified 59 times by D_X , and the corresponding spectrograms are misidentified 74 times by D_Y . The number of times that D_X and D_Y recognize the same error result is 9. A total of seven generated results are judged to be invalid because D_Y correctly identifies the spectrograms, but D_X incorrectly identifies the ground images. Therefore, out of the 700 experiments, there are 626 valid experiments, and 628 results are outputted. The results show that the double-check strategy reduces the error generated results by 87.8% (65/74) at the cost of losing 1.1% (7/626) valid results.

The messy ground images are misidentified 151 times by D_X , and the corresponding spectrograms are misidentified 166 times by D_Y . The number of times that D_X and D_Y

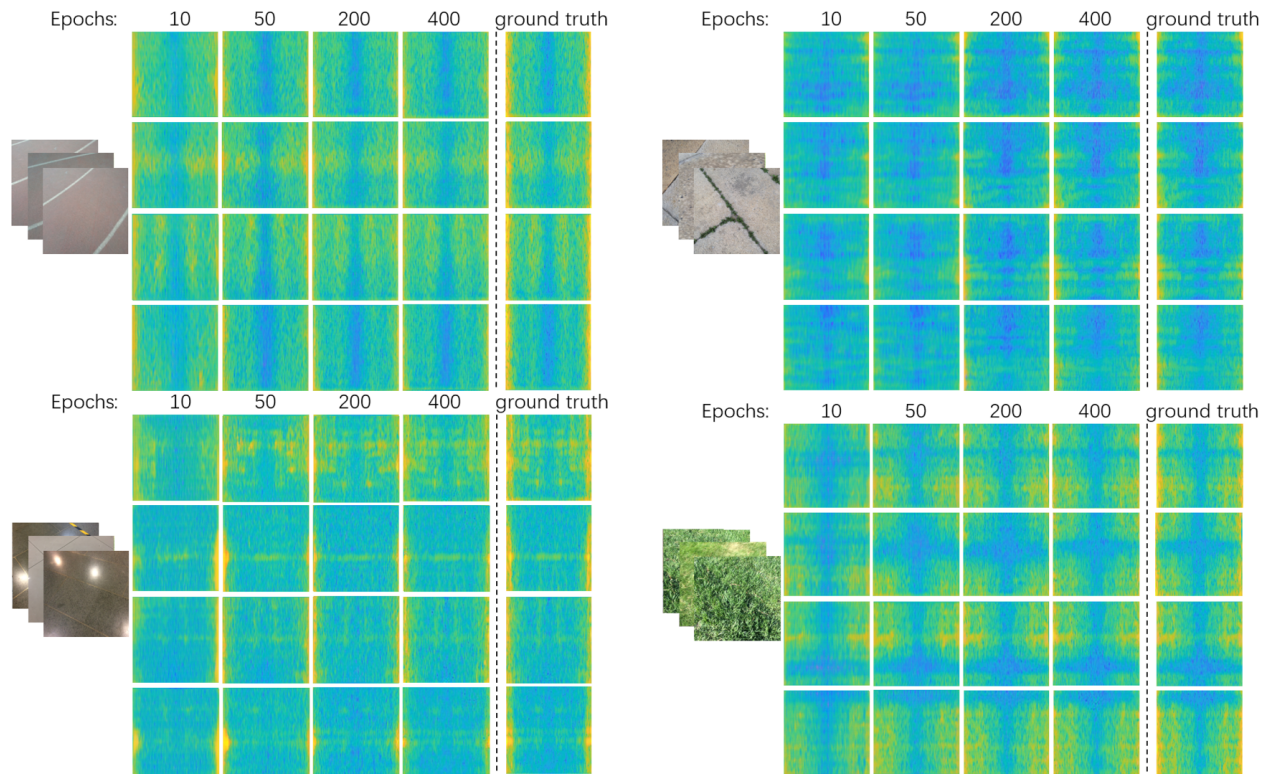


Fig. 6. Result of generated spectrogram with different epochs according to each category.

recognize the same error result is 25. A total of 11 generated results are judged to be invalid because D_Y correctly identifies the spectrograms, but D_X incorrectly identifies the ground images. Therefore, out of the 700 experiments, there are 534 valid experiments, and 548 results are outputted. The results show that the double-check strategy reduces the error generated results by 84.9% (141/166) at the cost of losing 2.1% (11/534) valid results.

VI. USER STUDY

A. System Design

Our system consists of a camera, a red button, a power amplifier, a vibrator (ALPS Inc., HAPTIC Reactor),¹ and a processor that are all integrated into a white cane (see Fig. 7). The main parts are concentrated in the upper part. The power supply and amplifier are stored in a backpack that can be easily carried by the user.

The vibrator is of $9 \times 10 \times 22.6$ mm and is fixed on the top of the cane and is easy to be sensed by the hand. It is a vibrational feedback device with two axes of vibrational freedom and has high vibrational freedom, enabling very realistic recreation of sensations even with complex waveforms. In addition, the touch sensations can be felt from a wide range of frequencies. The vibration response time is under 1 ms.

For practical use, the user may hold the handle of the cane. If he hopes to know the terrain information, he may click the red button to trigger the camera to capture the ground

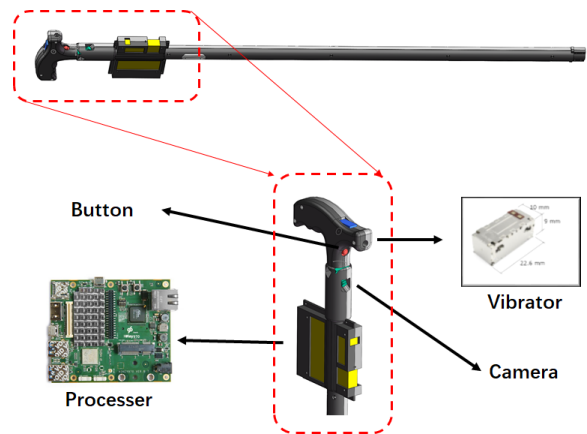


Fig. 7. Main component of the image-to-tactile cross-modal transformer.

image. Then, the image is fed into the processor for image-to-tactile signal transformation. After the signal transformation is completed, the vibrator will feedback the user's hand the tactile signal. In this way, the user may feel the ground terrain information. The most challenging problem is how to transform the image into the tactile signal.

B. Tests of Signal Generation

We further evaluate the performance of the developed prototype in real scenarios [43]. First, we test whether if the perceptually realistic vibrotactile signal could be generated and if the users could distinguish the category of the vibration at the same time. Then, another user study is conducted

¹<https://www.alps.com/prod/info/E/HTML/Haptic/>

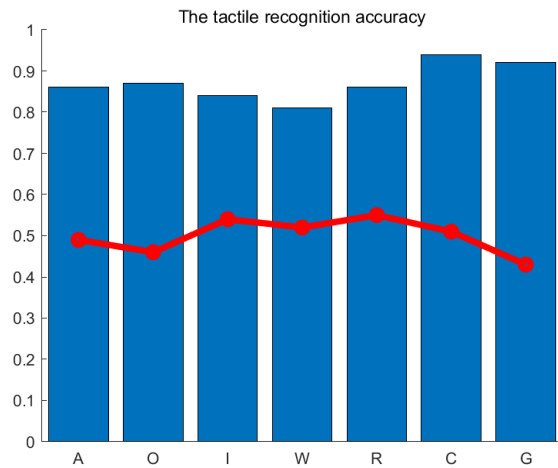


Fig. 8. Tactile recognition accuracy. The red line represents the percentage of the correctly recognized stimulus.

to evaluate the number of attempts required to simulate a blind user correctly identifying ground information in a real-world scenario. We invited ten volunteers who age from 21 to 26 (seven males and three females) to be involved in these evaluations.

During the test, participants are blindfolded and hold our model with their dominant hand. The researchers select categories and the source of vibration (ground truth or generated) based on the options on the test interface. After the vibration is generated, the users will answer three questions: what kind of touch they felt, whether this vibration is real or generated, and *how much realism did you feel*. For the last question, a visual analog scale (VAS) rating [44] is developed to describe the degree of realism for each stimulus. Participants rate the realism that they felt on an analog scale in this testing method and answer this question by observing and remembering a nine-bisected line segment before the experiment, where 0 means that the vibrations they felt are random and unreal and 10 means that the vibrations are extremely close to the real feel. Every participant tests five real and five generated vibrations in each category. As a result, a total of 700 meaningful test results are recorded. To avoid the sequential effects, we randomly assign the presentation order of these categories.

As shown in Fig. 8, the average recognition accuracy is 87.1%. These results indicate that the vibration signals are sufficiently distinguishable to allow the users to identify the ground categories.

In Fig. 8, we also show the correct answer rate with red lines, which is defined as the percentage of the correctly recognized stimulus. We observe that the results are rather close to 50%. This illustrates that most of the volunteers cannot distinguish the real tactile signals from the generated ones. In fact, all participants indeed admit that they do not find a significant difference between the real tactile signals and the generated ones. Such results reflect that the developed system is able to generate high-fidelity vibrotactile signals.

Furthermore, we list the feeling results in Fig. 9, which shows the results of how much realism the volunteers felt for each category. It is obvious that there is no significant

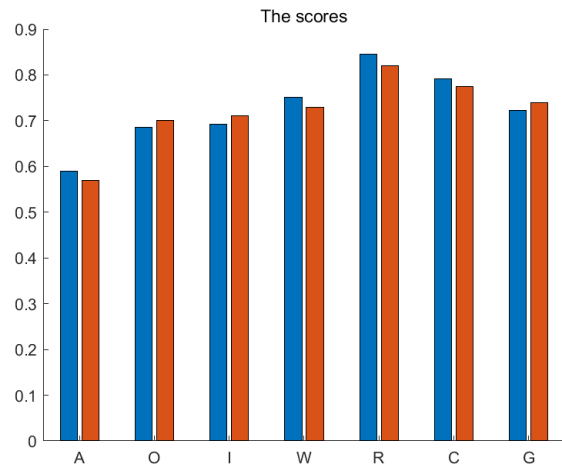


Fig. 9. Results of how much realism the volunteers feel for each category.

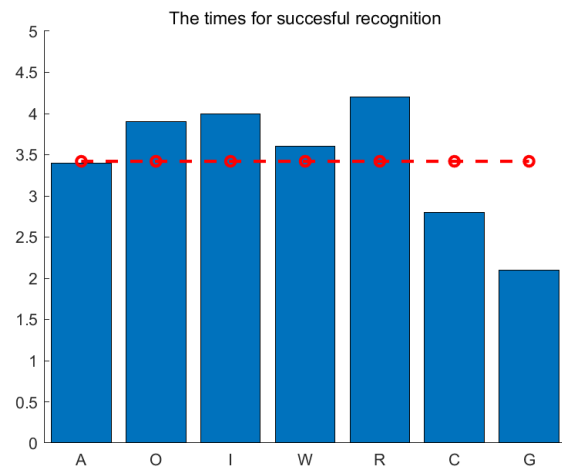


Fig. 10. Average numbers for successful recognition. The dashed line represents the mean number.

difference between the scores of generated vibration signals and real ones.

From the above-mentioned result, we find that the recognition accuracy of grass and carpet is relatively high, but their authenticity score is the lowest. On the contrary, the outdoor slab, which achieves the middle level of recognition performance, obtains the highest realistic score.

C. Tests of Real Scene Simulation

We blindfold all the volunteers and ask them to wear noise-isolating earplugs so that they could not hear the noise generated by dynamic obstacles in the environment. Thus, they could not judge the ground by association. In each scenario, participants are given enough room to move and to ensure the safety of the volunteers, and participants in each group of experiments are instructed by the researchers. One more thing to consider is that because participants with normal vision who are brought into a certain scene have a memory effect that seriously affects the accuracy of the test, they have to be dragged a long way by researchers into an unfamiliar environment to start the experiment. The participants hold

our model, take images of the ground, and then answer the results of their recognition based on the tactile vibrations they felt. When the correct judgment is made, the experiment is terminated and the researchers record the number of times they answered.

The results of real scene simulation experiments are shown in Fig. 10, with an average number of times of 3.4, in which the number of attempts required to correctly identify the grass is the least. This is because the tactile sensation of the grass is easy to feel. This conclusion is consistent with the previous experimental results.

VII. CONCLUSION

This article presents a portable device that provides tactile recognition assistance for visually impaired people. To train the model, we build a cross-modal data set and convert the time series signals into 2-D images for network processing. We train the generating network of vibration signals with the thought of GANs. In addition, we use double recognition constraints to greatly reduce the output of error generated results. Experimental results show that the developed system helps visually impaired people sense the ground and bring a better traveling experience for them. This article shows an important step toward the development of a more comprehensive assisting technology for visually impaired people. In the future, we will investigate how to combine the glasses and the white cane to construct more effective assisting technology.

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